

# Trajectory-Resolved CTMT Batteries:

A Conservative Theorem Ladder for Gauge-Protected Coherence,  
Operational Speed Identification, and Action-Scale Recovery on Tested Classes

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## Abstract

We present a conservative CTMT theorem package based on trajectory-resolved admissibility gates evaluated on synthetic tested classes. The central refinement is a Fisher-resolved trajectory gate: motion along weak or null Fisher directions is no longer classified as a physical violation by default, but is instead reported as gauge/null drift unless accompanied by an independent structural failure signature. This removes a known ambiguity of raw-trajectory gating and sharpens the separation between observationally resolved inconsistency and unresolvable parameter drift.

Three synthetic batteries are studied. First, a coil battery tests local chart admissibility, gauge protection, and structural separation on overlapping windows. Second, a pulse-arrival

propagation battery provides a tested-class operational route to a finite speed constant  $c_*$ . Third, a multi-route metrology battery provides a tested-class operational route to an action scale  $S_*$ . All results are synthetic and therefore validate pipeline logic, theorem wording discipline, and falsification structure rather than Nature directly.

The strongest currently defensible objects supported by the present battery are therefore: trajectory-resolved local charts with gauge-protected continuation on the tested coil class; a tested-class operational speed-identification theorem on the synthetic propagation class; and a tested-class operational action-scale recovery theorem on synthetic metrology routes. No global source-response closure theorem, no universal threshold law, and no empirical derivation of physical constants from laboratory data is claimed in this note.

## Contents

### 1 Purpose and claim stratification

This note records the strongest currently defensible theorem shape supported by the present synthetic battery. We distinguish the following claim classes.

- (C1) **Established local inputs.** Fisher-information geometry, least-squares inference, spectral conditioning logic, and admissible coarse-graining principles imported from the classical statistical literature and earlier CTMT notes.
- (C2) **Tested-class empirical theorems.** Statements supported on the synthetic coil, propagation, and metrology classes used here.
- (C3) **Operational bridge propositions.** Statements that connect local-chart admissibility to speed or action-scale identification on designed synthetic classes, but are not promoted to universal laws.
- (C4) **Explicit non-claims.** No theorem here asserts a global path-independent bundle theorem, a universal derivation of  $c$  or  $h$ , or laboratory closure of the final CTMT bridge.

### 2 Scope and non-claims

The operational meaning of the present battery is deliberately narrow.

1. All runs below use synthetic generated data. They test pipeline logic, theorem wording discipline, and falsification gates; they are not laboratory evidence.
2. The coil battery is a local-chart admissibility and persistence test. It does not by itself derive a universal speed constant and does not by itself derive quantization.
3. The propagation battery supplies an operational synthetic route to a finite speed constant on designed pulse-arrival data. It does not prove that CTMT has empirically derived the physical speed of light.
4. The action-scale battery supplies an operational synthetic route to  $S_*$  on designed metrology data. It does not constitute a new empirical remeasurement of Planck's constant.
5. No theorem below claims universal source-response closure, a path-independent global bundle theorem, or a derivation of new constants from the final public CTMT formulation.

### 3 Imported local inputs and admissibility spine

The present note uses the standard local CTMT observable model

$$y = y_0 + J \Delta\theta + u + \varepsilon, \quad (1)$$

where  $J$  is the local Jacobian,  $u$  denotes unresolved transport, and  $\varepsilon$  denotes measurement noise. The tested-class batteries use the following admissibility spine:

1. finite Jacobians and well-defined local Fisher matrices,
2. effective full rank on the stated resolved subspace,
3. bounded conditioning on the retained sector,
4. support confinement of unresolved covariance outside the resolved observational sector,
5. coarse-graining monotonicity on accepted windows,
6. consistency of overlapping local windows,
7. refusal under persistent rupture-like residual signatures.

For each accepted window  $k$ , let

$$F_k = J_k^\top C_k^{-1} J_k \quad (2)$$

be the Fisher matrix. The present refinement does not alter the role of Fisher geometry as the resolved distinguishability structure; it alters only the interpretation of motion in the weak or null Fisher sector.

### 4 Falsification logic

The program is counted as *falsified on the tested class* if any of the following signatures persist after stabilization and admissibility filtering.

- (F1) **Coverage collapse.** Accepted baseline-like trajectories exhibit persistent observational coverage failure.
- (F2) **Window inconsistency.** Overlapping accepted windows assign incompatible local chart structure without a corresponding admissibility failure.
- (F3) **Rupture-like residual behavior.** Residual evolution develops persistent discontinuity-like signatures while the trajectory is still scored benign.
- (F4) **Blockwise mismatch.** Axial/radial or analogous sector decomposition exhibits persistent mismatch while the trajectory is still scored benign.
- (F5) **Resolved-trajectory ambiguity.** The resolved Fisher gate fails to separate benign tested scenarios from injected structural violations.
- (F6) **Bridge failure.** The propagation battery fails to recover a stable finite speed on admissible synthetic transport, or the metrology routes fail to recover a stable action scale across independent synthetic probes.

**Proposition 1** (Tested-class falsifiability). *If any of the falsifiers (F1)–(F6) occurs persistently on stabilized admissible windows or tested synthetic bridge classes, then the present trajectory-resolved CTMT package fails on the tested class and must be weakened, restricted, or abandoned in its present form.*

**Proof 1.** Each falsifier contradicts one of the essential claims made in this note: admissibility stability, structural separation, gauge protection, finite-speed transport identification, or independent-route action-scale recovery. Persistent occurrence of any falsifier therefore contradicts the tested-class framework itself rather than a merely decorative implementation detail.

## 5 Trajectory-resolved Fisher gate

Let  $F_k = U_k \Lambda_k U_k^\top$  be the Fisher matrix in window  $k$ , and decompose its eigenspaces into a resolved sector  $U_{r,k}$  and a weak/null sector  $U_{s,k}$ . For the local increment

$$\Delta\theta_k = \theta_{k+1} - \theta_k,$$

define the resolved step energy

$$d_{r,k}^2 = \Delta\theta_k^\top U_{r,k} \Lambda_{r,k} U_{r,k}^\top \Delta\theta_k, \quad (3)$$

and the weak/null drift magnitude

$$d_{s,k} = \|U_{s,k}^\top \Delta\theta_k\|. \quad (4)$$

The present trajectory-resolved gate judges physical inconsistency from the resolved series  $d_{r,k}$  together with structural diagnostics, while  $d_{s,k}$  is reported as gauge/null drift unless an independent structural failure signature is present.

**Definition 2** (Independent structural failure signature). A window chain is said to exhibit an independent structural failure signature if at least one of the following persists above the tested threshold: coverage collapse, severe block mismatch, rupture-like residual behavior, overlap inconsistency, or resolved-subspace trajectory inconsistency.

**Theorem 1** (Fisher-resolved gauge protection on the tested class). *Assume a local coherent chart with full resolved Fisher rank, bounded conditioning on the stated effective subspace, negligible leakage into the resolved covariance sector, and small normalized block gap. Then motion concentrated in weak/null Fisher directions is classified by the present battery as gauge/null drift rather than structural violation unless an independent structural failure signature is present.*

**Proof 3.** The gate acts on the resolved Fisher geometry rather than on raw parameter motion. By construction, motion orthogonal to the resolved eigenspace contributes negligibly to the resolved quadratic variation  $d_{r,k}$  while remaining visible in the null-drift observable  $d_{s,k}$ . Hence null-aligned parameter motion is not, by itself, observational evidence of structural breakdown. A violation verdict therefore requires one of two things: either a persistent excess in the resolved subspace inconsistency series, or a separate structural signature such as severe coverage loss, block mismatch, rupture-like behavior, or overlap inconsistency. The gate is therefore aligned with the observational geometry encoded by the Fisher metric: only motion seen in the resolved sector, or independently supported structural failure, is promoted to a physical inconsistency verdict.

**Corollary 1** (Gauge-resolved separation). *On the tested class, null-supported drift alone does not imply physical inconsistency. Structural refusal requires resolved inconsistency or an independent failure signature.*

**Proof 4.** This is the direct consequence of the theorem: null-supported motion is explicitly tracked but not promoted to structural violation in the absence of an independent refusal trigger.

## 6 Synthetic coil battery

The synthetic coil battery used one loop of radius  $R = 5$  cm, 60 axial plus 40 radial sensors, 12 windows, and synthetic field noise  $\sigma_B = 2 \times 10^{-8}$  T. The local anchor was

$$\theta_{\text{anchor}} = (4.0, 5.8 \times 10^{-4}, -4.2 \times 10^{-4}).$$

The anchor Jacobian singular values were approximately

$$s(J_{\text{anchor}}) \approx (1.6301, 1.7373 \times 10^{-2}, 1.6365 \times 10^{-4}),$$

so the local chart is full-rank but badly conditioned, which is precisely why local anchoring and effective-subspace reasoning matter.

### Trajectory summary on the tested coil class

| Scenario            | Grade              | Coverage med. | Resolved traj. | Block gap | Consistency | Accepted windows |
|---------------------|--------------------|---------------|----------------|-----------|-------------|------------------|
| baseline            | SUPPORT            | 0.935         | 1.710          | 0.070     | 0.093       | 3                |
| anisotropy          | STRONG             | 0.950         | 3.085          | 0.094     | 0.085       | 9                |
| radius coupling     | DETECTED VIOLATION | 0.895         | 2.320          | 1.328     | 0.529       | 1                |
| dispersion          | SUPPORT            | 0.925         | 1.743          | 0.071     | 0.075       | 4                |
| null seepage        | SUPPORT            | 0.930         | 2.982          | 0.080     | 0.055       | 4                |
| causal violation    | DETECTED VIOLATION | 0.020         | 4.585          | 0.493     | 5.371       | 0                |
| unit leak           | DETECTED VIOLATION | 0.000         | 1.534          | 0.550     | 0.168       | 0                |
| transport reuse     | SUPPORT            | 0.925         | 2.230          | 0.053     | 0.055       | 1                |
| weak activation toy | STRONG             | 0.980         | 1.642          | 0.061     | 0.084       | 12               |

Table 1: Trajectory-resolved coil battery summary on the tested synthetic class. “Resolved traj.” denotes the resolved-subspace trajectory inconsistency statistic. “Consistency” denotes the aggregate window-consistency statistic.

**Proposition 2** (Synthetic separation on the tested coil class). *On the tested synthetic coil family, the trajectory-resolved battery separates gauge-compatible or benign scenarios from injected structural violations more cleanly than the earlier raw-trajectory gate. In particular, the present run supports the benign or gauge side for baseline, anisotropy, dispersion, null seepage, transport reuse, and weak activation, while refusing radius coupling, causal violation, and unit leak.*

**Proof 5.** Table 1 shows that the baseline scenario is retained on the benign side with coverage median 0.935, resolved trajectory statistic 1.710, normalized block gap 0.070, and aggregate consistency statistic 0.093. The null seepage case is likewise retained, with coverage median 0.930, block gap 0.080, and consistency statistic 0.055. This is exactly the class for which the gauge-protection refinement matters: weak/null-supported motion is present, but the structural diagnostics remain inside the tested admissibility regime.

By contrast, the radius-coupling scenario is refused because its normalized block gap rises to 1.328 and its consistency statistic to 0.529, which is structurally incompatible with benign reuse of the same resolved chart. The causal-violation scenario is refused even more strongly: its coverage median collapses to 0.020, its resolved trajectory statistic rises to 4.585, and its consistency statistic rises to 5.371. The unit-leak case is refused through complete coverage collapse and a large block gap of 0.550. Therefore the tested synthetic class is separated by the present battery into benign/gauge-compatible and detected-violation families.

**Corollary 2** (Necessity of the trajectory-resolved gate on the tested coil class). *A raw parameter-trajectory gate is not sufficient as a general detector of physical inconsistency on the tested coil class. Resolved Fisher geometry is required in order to distinguish null-supported gauge drift from structural failure.*

**Proof 6.** The battery retains null seepage and related benign scenarios while still refusing explicitly structural violations. This separation depends on analyzing the resolved Fisher sector rather than raw parameter motion. Therefore the tested result itself provides a counterexample to any rule that would classify all large parameter motion as structural failure.

**Remark 1.** The coil battery sharpens the causality bridge by protecting null/gauge motion from being misread as structural violation. It still does not by itself identify a universal propagation speed from static coil data.

## 7 Synthetic propagation battery for a speed constant $c_*$

A separate pulse-arrival battery was used to test the speed layer that static coil data cannot close alone. Sensor positions  $x_i$  lay on a line and arrival times were generated by

$$t_i = t_0 + \frac{x_i}{c_*} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_t^2),$$

with violating scenarios given by a superluminal-like law, a noncausal shuffle, and a multipath-like discontinuity.

### Propagation results on the tested class

| Scenario            | Grade              | $\hat{c}_*$ [m/s] | Relative error | Timing RMS [ps] |
|---------------------|--------------------|-------------------|----------------|-----------------|
| baseline            | STRONG             | 299938196.977     | 4.861e-04      | 19.877          |
| benign dispersion   | STRONG             | 299822165.405     | 9.909e-05      | 21.211          |
| superluminal like   | DETECTED VIOLATION | 353803600.290     | 1.802e-01      | 19.126          |
| noncausal shuffle   | DETECTED VIOLATION | 875044863.710     | 1.919e+00      | 10850.430       |
| multipath violation | DETECTED VIOLATION | 304235074.487     | 1.482e-02      | 101.761         |

Table 2: Synthetic propagation battery results for operational speed identification on the tested class.

**Theorem 2** (Synthetic operational speed identification on the tested class). *On the tested synthetic pulse-arrival class, the propagation battery recovers a stable operational speed constant  $\hat{c}_*$  numerically close to the injected constant in admissible scenarios and refuses noncausal or structurally violating scenarios.*

**Proof 7.** Table 2 shows that the baseline synthetic pulse family is accepted with fitted speed

$$\hat{c}_* = 299938196.977 \text{ m/s},$$

relative error  $4.861 \times 10^{-4}$ , and timing RMS 19.877 ps. The benign-dispersion family is likewise accepted with fitted speed

$$\hat{c}_* = 299822165.405 \text{ m/s},$$

relative error  $9.909 \times 10^{-5}$ , and timing RMS 21.211 ps. These scenarios therefore satisfy the tested-class requirement of stable finite-speed recovery under admissible perturbation.

By contrast, the superluminal-like scenario is refused with a relative error of  $1.802 \times 10^{-1}$ , the noncausal shuffle is refused with an extreme fit

$$\hat{c}_* = 875044863.710 \text{ m/s}$$

and timing RMS 10850.430 ps, and the multipath violation is refused despite a smaller relative error because its timing residual structure is not compatible with a single admissible propagation law. The battery therefore separates admissible finite-speed transport from explicit violating constructions on the tested class.

**Corollary 3** (Current status of the speed bridge). *The synthetic results support a tested-class program from CTMT admissibility to finite-speed operational transport: the coil battery cleans the local chart and gauge logic, while the propagation battery supplies the explicit speed layer. No final empirical derivation of the physical speed of light from laboratory data is claimed here.*

**Proof 8.** The theorem above establishes only tested-class operational recovery on designed synthetic transport. Since the data are synthetic rather than laboratory measurements, the bridge is operational and tested-class only.

## 8 Synthetic action-scale battery for $S_*$ and quantization

Three synthetic metrology routes were used:

1. blackbody radiometry via synthetic Planck-spectrum fitting,
2. Josephson voltage–frequency metrology,
3. a toy quantum FDT-style crossover route.

The target action scale was the SI Planck constant

$$h = 6.62607015 \times 10^{-34} \text{ J s}.$$

### Action-scale results on the tested class

| Route           | $\hat{S}_*$ [J s] | Relative error | Accepted |
|-----------------|-------------------|----------------|----------|
| blackbody       | 6.62573843e-34    | 5.006e-05      | True     |
| josephson       | 6.62543247e-34    | 9.624e-05      | True     |
| quantum fdt toy | 6.61683860e-34    | 1.393e-03      | True     |

Table 3: Synthetic action-scale identification across independent metrology routes.

**Theorem 3** (Synthetic operational action-scale identification on the tested class). *On the tested synthetic metrology routes, the operational pipeline estimates an action scale  $\hat{S}_*$  numerically equal to the injected constant  $h$  within the stated synthetic tolerances.*

**Proof 9.** Table 3 shows that the blackbody route returns

$$\hat{S}_*^{(\text{bb})} = 6.62573843 \times 10^{-34} \text{ J s},$$

with relative error  $5.006 \times 10^{-5}$ . The Josephson route returns

$$\hat{S}_*^{(\text{J})} = 6.62543247 \times 10^{-34} \text{ J s},$$

with relative error  $9.624 \times 10^{-5}$ . The toy FDT route returns

$$\hat{S}_*^{(\text{FDT})} = 6.61683860 \times 10^{-34} \text{ J s},$$

with relative error  $1.393 \times 10^{-3}$ . All three routes satisfy their synthetic acceptance gates. Therefore the tested-class operational pipeline numerically recovers the injected action scale across independent synthetic probes.

**Corollary 4** (Current status of the quantization bridge). *The synthetic results support the operational statement that a CTMT-style metrology pipeline can recover a stable action scale on tested synthetic routes. They do not, by themselves, constitute new empirical evidence beyond the design of the synthetic experiment.*

**Proof 10.** The theorem above establishes consistency of independent synthetic identification routes with the injected constant. Since the routes are synthetic by design, the conclusion is operational and tested-class only, not an empirical remeasurement claim.

## 9 Operational domain

All theorem-level and proposition-level statements in this note are restricted to the following operational domain.

- (D1) The Fisher trajectory is built from stabilized local windows with a well-defined resolved subspace.
- (D2) Conditioning is evaluated on the effective retained sector and remains bounded there.
- (D3) Blockwise mismatch, rupture metrics, and overlap consistency are treated as empirical protocol diagnostics rather than universal first-principles invariants.
- (D4) The propagation and metrology batteries are synthetic designed experiments used only to test operational bridge logic.
- (D5) Stability under arbitrary kernels, arbitrary threshold choices, or laboratory deployment is not claimed here.

**Proposition 3** (Scope fencing). *All theorem-level and proposition-level statements in the present note are valid only inside the operational domain (D1)–(D5).*

**Proof 11.** Outside the operational domain, the stabilized resolved subspace, the tested thresholds, or even the meaning of the synthetic operational route may fail to be well-defined in the present sense. The claims proved here are therefore explicitly fenced to the stated operational regime.

## 10 Final theorem ladder

**Corollary 5** (Final supported theorem shape). *The strongest theorem ladder supported by the present battery is:*

1. *a Fisher-resolved gauge-protection theorem on the tested local chart class;*
2. *a tested-class separation proposition for benign/gauge scenarios versus injected structural violations on the synthetic coil class;*
3. *a tested-class operational speed-identification theorem on the synthetic propagation class;*
4. *a tested-class operational action-scale-identification theorem on independent synthetic metrology routes.*

**Proof 12.** Item (1) is provided by the gauge-protection theorem in the trajectory-resolved Fisher section. Item (2) is established by the coil-battery separation proposition and the summary in Table 1. Item (3) is established by the propagation theorem and Table 2. Item (4) is established by the action-scale theorem and Table 3. No stronger theorem—such as a universal constant derivation or global CTMT closure theorem—is supported by the present synthetic evidence.

## 11 Explicit non-claims

The following are not claimed in this paper.

1. No universal laboratory derivation of  $c$  is claimed here.
2. No universal laboratory derivation of  $h$  is claimed here.



3. No claim is made that the tested thresholds are universal across all CTMT kernels.
4. No claim is made that the static coil battery alone completes a full source-response or SI-calibration theorem.
5. No path-independent global bundle theorem is established here.
6. No synthetic battery is treated as laboratory evidence.

## 12 Operational falsifications going forward

The present package should be weakened or revised if any of the following occurs persistently under future testing.

1. Baseline or null-supported gauge scenarios continue to fail after resolved Fisher gating.
2. Resolved-subspace inconsistency remains unable to distinguish benign from violating families.
3. Propagation batteries fail to recover a stable finite speed on admissible designed transport.
4. Independent metrology routes fail to return a stable  $S_*$  across action-scale probes.
5. Support collapses under overlap continuation or transport reuse.

## 13 Conclusion

The present refinement materially improves the CTMT bridge program by aligning trajectory refusal logic with resolved Fisher geometry and by treating weak/null Fisher motion as gauge-like unless an independent structural failure is present. This sharpens local-chart admissibility on the tested synthetic coil class and removes a known ambiguity of raw-trajectory gating. A separate synthetic propagation battery then supplies the explicit speed layer, while an independent synthetic metrology battery supplies the explicit action-scale layer.

The resulting theorem package is intentionally conservative. It supports tested-class local gauge protection, tested-class structural separation, tested-class finite-speed operational identification, and tested-class action-scale recovery. It does not claim laboratory closure, universal threshold laws, or derivation of physical constants from Nature.

## A Python companion and reproducibility

A synthetic reproducibility companion is provided in the script `The uploaded companion script` is organized around three tested-class batteries: a coil battery with Fisher-resolved trajectory gating, a propagation battery for operational speed identification, and an action-scale battery for synthetic metrology recovery; it is also designed to emit structured summary artifacts including CSV, JSON, text, and image outputs.

```

#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""
CIMT v6 final synthetic battery-attack companion
=====

This script is a synthetic-only companion for the CIMT battery-attack appendix.
It implements three tested-class batteries:

1) Coil v6 battery (Fisher-resolved trajectory gate)
2) Synthetic propagation battery for an operational speed constant c*
3) Synthetic action-scale battery for an operational action constant S*

Important:
- All data generated by this script are synthetic.
- The outputs test pipeline logic, theorem wording discipline, and falsification gates.
- The script does NOT provide empirical laboratory evidence and does NOT claim derivation
  of physical constants from Nature.

Outputs generated in the current working directory:
- cmt.v6.coil.summary.csv
- cmt.v6.coil.report.json
- cmt.v6.coil.dashboard.png
- cmt.speed.battery.summary.csv
- cmt.speed.battery.report.json
- cmt.quant.battery.summary.csv
- cmt.quant.battery.report.json
- cmt.v6.combined.summary.json
- cmt.v6.run.summary.txt
"""

from __future__ import annotations

import json
from dataclasses import dataclass, asdict
from typing import Tuple, Dict, List

import numpy as np
import numpy.linalg as npl
import pandas as pd
import matplotlib.pyplot as plt

# =====
# Global physical constants used by the synthetic batteries
# =====
MU0 = 4e-7 * np.pi
C.TRUE = 299.792.458.0
H.TRUE = 6.62607015e-34
K.B = 1.380649e-23
C.LIGHT = 299.792.458.0
E.CHARGE = 1.602176634e-19

# =====
# Coil v6 battery
# =====
@dataclass
class CoilConfig:
    radius_m: float = 0.05
    nseg: int = 500
    nz: int = 60
    nr: int = 40
    zmax: float = 0.06
    rmax: float = 0.08
    sigma_B: float = 2e-8
    theta_true: Tuple[float, float, float] = (4.0, 6e-4, -4e-4)
    theta_anchor: Tuple[float, float, float] = (4.0, 5.8e-4, -4.2e-4)
    nwindows: int = 12
    lam_cov: float = 0.35
    rho_cov: float = 1e-15
    rupture_limit: float = 3.5
    seed: int = 7

def loop_polyline(radius: float = 0.05, nseg: int = 500, center=(0.0, 0.0, 0.0)) -> np.ndarray:
    phi = np.linspace(0, 2 * np.pi, nseg, endpoint=False)
    return np.c_[
        radius * np.cos(phi) + center[0],
        radius * np.sin(phi) + center[1],
        np.zeros_like(phi) + center[2],
    ]

def axial_and_radial_sensors(nz: int = 60, zmax: float = 0.06, nr: int = 40, rmax: float = 0.08) -> np.ndarray:
    z_axis = np.linspace(-zmax, zmax, nz)
    axis_pts = np.c_[np.zeros(nz), np.zeros(nz), z_axis]
    r_line = np.linspace(0.0, rmax, nr)
    rad_pts = np.c_[r_line, np.zeros(nr), np.zeros(nr)]
    return np.vstack([axis_pts, rad_pts])

def biot_savart_line(sensors: np.ndarray, polyline: np.ndarray, current: float = 1.0) -> np.ndarray:
    p0 = polyline
    p1 = np.roll(polyline, -1, axis=0)
    dL = p1 - p0
    B = np.zeros((sensors.shape[0], 3))
    for i in range(sensors.shape[0]):
        r = sensors[i][None, :] - p0
        rn = npl.linalg.norm(r, axis=1) + 1e-12
        rh = r / rn[:, None]
        cross = np.cross(dL, rh)
        contrib = MU0 * current * cross / (4 * np.pi * rn[:, None] ** 2)
        B[i, :] = np.sum(contrib, axis=0)
    return B

def biot_savart_multi(sensors: np.ndarray, conductors: List[np.ndarray], currents: List[float]) -> np.ndarray:
    B = np.zeros((sensors.shape[0], 3))
    for poly, I in zip(conductors, currents):
        B += biot_savart_line(sensors, poly, I)
    return B

def forward_Bz(theta: Tuple[float, float, float], sensors: np.ndarray, conductors: List[np.ndarray]) -> np.ndarray:
    I, dx, dy = map(float, theta)
    shifted = [poly + np.array([dx, dy, 0.0]) for poly in conductors]
    return biot_savart_multi(sensors, shifted, [I] * len(conductors))[:, 2]

```

The role of the Python companion in the present paper is purely operational:

1. construct the synthetic scenario families used in the tested classes;
2. compute the local Fisher objects and trajectory diagnostics;
3. execute the propagation and action-scale bridge batteries;
4. generate machine-readable summaries for reproducibility and audit.

The script is not itself a theorem. Its role is to accompany the tested-class empirical statements of this note by making the operational pipeline explicit and reproducible.

## References

- [1] R. A. Fisher. Theory of Statistical Estimation. *Mathematical Proceedings of the Cambridge Philosophical Society*, 22(5):700–725, 1925.
- [2] N. N. Cencov. *Statistical Decision Rules and Optimal Inference*. American Mathematical Society, Translations of Mathematical Monographs, Vol. 53, 1982.
- [3] C. R. Rao. *Linear Statistical Inference and Its Applications*. Wiley, 2nd edition, 2001.
- [4] M. Planck. Über das Gesetz der Energieverteilung im Normalspectrum. *Annalen der Physik*, 4:553–563, 1901.
- [5] B. D. Josephson. Possible new effects in superconductive tunnelling. *Physics Letters*, 1(7):251–253, 1962.
- [6] H. B. Callen and T. A. Welton. Irreversibility and generalized noise. *Physical Review*, 83(1):34–40, 1951.
- [7] A. Einstein. Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig? *Annalen der Physik*, 323(13):639–641, 1905.